our agnosticity to how the models are constructed, we view language models as natural language interfaces (see Lee et al., Forthcoming).

As Bommasani et al. (2021, §1.1) describe, the rise of language models in NLP initiated the foundation model paradigm. Specifically, ELMo (Peters et al., 2018), GPT (Radford et al., 2018), and BERT (Devlin et al., 2019) demonstrated that pretraining using language modeling objectives could produce powerful general-purpose representations for many downstream use cases, building on prior evidence of the successes of pretraining (Mikolov et al., 2013; Pennington et al., 2014). Further, these works, especially GPT and later GPT-2 (Radford et al., 2019), produced models with qualitatively better generative capabilities than what had been seen previously.

Together, these formative works ushered in significant change in the status of language modeling in NLP: language models rapidly became the substrate for almost all modeling work, especially with the advent of open infrastructure through Hugging Face Transformers (Wolf et al., 2019) and models developed for languages beyond English (e.g. multilingual-BERT, XLM; Devlin et al., 2019; Conneau & Lample, 2019). Since then, we have seen a proliferation of different organizations building language models, often through conceptually similar means, with a rapid growth in scale and resource-intensivity. Notably, some of the models (e.g. TNLG v2 (530B)) we benchmark 1000× larger than ELMo and BERT. These models can cost millions of dollars to train, requiring extensive systems-level optimizations and dedicated large-scale compute (Narayanan et al., 2021). These changes have also translated from research to deployment: language models are directly exposed as commercial APIs or are integrated into ubiquitous products (see Bommasani et al., 2022, §5) as part of an emerging commercial ecosystem (Bommasani et al., 2021, §1.2).[78]

Benchmarks in NLP. Similar to language modeling, benchmarking has a long history in NLP. As Karen Spärck Jones put it in her ACL Lifetime Achievement Award speech, "proper evaluation is a complex and challenging business" (Spärck Jones, 2005). To address this challenge, the practice of benchmarking rose to prominence as the core methodology in the 1980's and, especially, the 1990's (see Liberman, 2010; Spärck Jones & Galliers, 1995). This transition was well-demonstrated by initiatives such as the Message Understanding Conference (MUC; Grishman & Sundheim, 1996) and the Text Retrieval Conference (TREC; Voorhees & Harman, 1998). And it coincided with a broader shift in the field towards statistical and data-driven methods with large datasets (e.g. the Penn Treebank (Marcus et al., 1999)) and new venues like the Conference on Empirical Methods for Natural Language Processing (EMNLP, 1996).

More than a decade later, with the rise of deep learning in the 2010s (Collobert & Weston, 2008; Turian et al., 2010; Collobert et al., 2011; Socher et al., 2011a;b; Sutskever et al., 2011; Mikolov et al., 2013; Pennington et al., 2014; Sutskever et al., 2014; Bahdanau et al., 2015; Luong et al., 2015; Vaswani et al., 2017), larger benchmarks such as SNLI (Bowman et al., 2015) and SQuAD (Rajpurkar et al., 2016) were developed to provide both adequate data for training systems in addition to evaluating systems. This parallels concurrent developments in other areas of AI: most notably, the ImageNet benchmark (Deng et al., 2009) that shaped modern computer vision. Like their predecessors, these benchmarks assign each model a single score (e.g. the SQuAD F1 score) to measure the accuracy for a single task.

As more general-purpose approaches to NLP grew, often displacing more bespoke task-specific approaches, new benchmarks such as SentEval (Conneau & Kiela, 2018), DecaNLP (McCann et al., 2018), GLUE (Wang et al., 2019b), and SuperGLUE (Wang et al., 2019a) co-evolved to evaluate their capabilities. In contrast to the previous class of benchmarks, these benchmarks assign each model a vector of scores to measure the accuracy for a suite of scenarios. In some cases, these benchmarks also provide an aggregate score (e.g. the GLUE score, which is the average of the accuracies for each of the constituent scenarios).

More recently, this theme of meta-benchmarks that assess model accuracy across a range of tasks has continued (see Bommasani et al., 2021, §4.4.3): for example, GEM (Gehrmann et al., 2021) provides a suite for natural language generation tasks, XTREME (Hu et al., 2020) provides a suite for tasks spanning numerous languages, and GEMv2 (Gehrmann et al., 2022a) provides a suite for generation across languages.

[78] See the emerging startup ecosystem for foundation models: `https://www.scalevp.com/blog/introducing-the-scale-generative-ai-index`

This approach is also the dominant approach to language model evaluation,[79] often with even broader collections: Brown et al. (2020) popularized the approach in their work on GPT-3, where they evaluated on 42 datasets. Indeed, this is the approach used in all the works that introduced models we evaluate in this work. Efforts like the EleutherAI Language Model Evaluation Harness (Gao et al., 2021b), HuggingFace's Evaluate library (von Werra et al., 2022), and Big-Bench (Srivastava et al., 2022) have centralized and expanded these evaluations into systematic repositories.

Situated against this landscape, what differentiates our work is our holistic approach, which manifests in both our benchmark design process and our concrete benchmark. HELM is the byproduct of an explicit two-step process: we taxonomize the space for language model evaluation, structured around use cases (scenarios) and desiderata (metrics), and then systematically select points in a way that reflects our priorities. This makes explicit the aspiration, the concrete benchmark, and, consequently, what our benchmark lacks that we should aspire to evaluate. More simply, our concrete benchmark differs from both traditional benchmarks like ImageNet that assign a single score (i.e. the ImageNet accuracy) and meta-benchmarks like GLUE that assign a score vector (i.e. the accuracies on the GLUE datasets) to each model. Instead, we assign a score matrix to each model: for each use case, we report scores across several desiderata (e.g. accuracy, calibration, robustness, fairness, efficiency).

Independent of the fact we measure holistically, one may wonder what the relationship is between the scenarios we select and those evaluated in prior works. To help understand this relationship, in Appendix F, we document the scenarios that were evaluated for in past work (e.g. the scenarios evaluated by Chowdhery et al. (2022) in the PaLM paper or by Gao et al. (2021b) in the EleutherAI Language Model Evaluation Harness) as well as past results for the models we evaluate on our scenarios (e.g. the **HellaSwag** accuracy reported by Brown et al. (2020) in the GPT-3 paper). Further, to build on BIG-Bench specifically, we highlight that our codebase integrates all BIG-Bench scenarios, augmented with metrics beyond accuracy and the ability to evaluate all models we support. We emphasize that at this time, no common standard exists for language modeling evaluation, especially as the capabilities, harms, and limitations of these models are still being understood through the ongoing design of evaluations. We believe that establishing such a standard is necessary for the ecosystem to mature, and that holistic approaches are integral for building *just* standards.

10 What is missing

One of our three requirements for holistic evaluation is the recognition of limitations: holistic evaluation should foreground where what is implemented falls short of the ambition of evaluating models in their totality. Our benchmark foregrounds the limitations of our current benchmark by design: our benchmark is a subset of a pre-specified taxonomy. That is, the difference between what is in the taxonomy and what is in the benchmark identifies what we currently miss.

While it is useful to articulate what is missing, given the immense scale of the use cases and desiderata we could have for language models, we believe it is necessary to have clear **priorities** on how to navigate the space of what we lack. These priorities are deeply subjective: many compelling arguments can be presented to increase focus on any specific region of the design space for language models and language model evaluation. Indeed, this is also made clear through the diversity of different works happening in parallel throughout the AI community to evaluate language models.

In this section, having conducted our holistic evaluation, we reflect on what we believe should be prioritized based on this experience. That is, we identify specific regions of the language model evaluation design space which we hope the community will improve, either because we feel HELM currently only scratches the surface or because we believe these concepts have been largely neglected in the broader NLP and AI communities. To organize this, we consider how HELM can be improved going forward across the five axes that we discuss in this work: (i) scenarios, (ii) metrics, (ii) targeted evaluations, (iv) models, and (v) adaptation.

[79] We note Efrat et al. (2022) as a very recent counter-example to this trend that takes a much more minimalistic and succinct unit-testing perspective.

10.1 Missing scenarios

Since we define scenarios in terms of tasks, domains, and languages, we begin by considering what we miss for each of these. For tasks, we emphasize that we deliberately chose to prioritize classical user-facing tasks, but other tasks also can confer significant societal impact, even if they are not user-facing (e.g. syntactic parsing and natural language inference). Beyond this, even while our focus is oriented by societal impact in task selection, other tasks may be more discriminative in identifying specific properties, limitations, or capabilities of models (e.g. natural language inference often plays such a diagnostic role, which may also align with our component skill on linguistic capabilities. We also explicitly acknowledge certain tasks that we do not currently implement as an oversight in terms of determining what was user-facing, such as data-to-text generation which has many existing benchmarks (e.g. Gardent et al., 2017; Novikova et al., 2017).

We especially highlight how the deployment of language models is giving rise to entirely new tasks, beyond the standard purview of the NLP and AI research communities. For example, several startups such as Jasper.AI[80] and Copy.AI[81] are deploying systems for copy-writing with language models as their foundation.[82] Even further, we are seeing the rise of new types of creative and generative applications like story and email generation (see Lee et al., 2022b; Forthcoming). We believe it is incumbent on the NLP and AI research communities to develop evaluations for these new use cases, which not only may confer economic/commercial value but societal impact going forward, as key instances where practical deployments can inform research priorities.

For domains we highlight three priorities. First, for coverage of *what*, i.e. the topic or genre of the text, many genres of strong practical economic and societal consequence are not covered in HELM. As examples, we highlight biomedical and clinical data, financial data, education data, and customer service, pointing to how language technologies are increasingly widespread[83] in these sectors but we have no coverage of these anywhere in our benchmark. We do note we have some coverage of legal data via **LegalSupport**, though the realism of the entailment-based data for legal language technologies could stand to be improved (cf. Guha et al., 2022). Second, for coverage of *when*, i.e. the time period of the text, we highlight the contrast between the ever-changing nature of language, world, and society with the relative rigidity of many current language models in being able to update/edit knowledge (Lazaridou et al., 2021). Early efforts to create such evaluations exist such as StreamingQA (Liska et al., 2022). And symmetrically, while perhaps of less commercial utility, we have seen language models being used for inquiry in the computational social sciences and digital humanities, where evaluation of performance on historic/ancient texts may be especially pertinent (see Bamman & Burns, 2020; Yamshchikov et al., 2022). Third, for coverage of *who*, we note that there are many standard demographic categories within the US where we currently have no evaluation (e.g. age, socioeconomic status), and in fact the metadata (i.e. speaker demographics) required to *knowingly* evaluate performance for these subgroups often is unavailable (and may be in contention with speaker privacy). Beyond this, we highlight two further nuances that relate to speaker identity: (i) native vs. non-native speakers of English as well as (ii) speaker demographics that are culturally situated in English-speaking contexts beyond the US (e.g. caste in India as discussed by Sambasivan et al. (2021) and Bhatt et al. (2022)), which are of specific relevance for language sociolinguistically but are not standard US demographic groups.

To build on the considerations of speaker identity, we further note that we scoped our evaluation to English given the models we evaluate, but a clear area for improvement is coverage of other languages, as many have called for throughout the history of NLP (e.g. Bender, 2009; 2011; 2012; Joshi et al., 2020). Currently, we believe we take important strides in coverage of English varieties both in terms of varieties like African American English (mainly through **TwitterAAE** but also through data augmentations following Ziems et al. (2022)) and English spoken in different nations through **ICE**. But we emphasize that a specific place to improve is to situate these evaluations in the context of societally consequential use cases, whereas currently to ensure some coverage we are relegated to measuring performance for these varieties in language modeling

[80] `https://www.jasper.ai/`
[81] https://www.copy.ai/
[82] See `https://www.scalevp.com/blog/introducing-the-scale-generative-ai-index` for more examples.
[83] For example, see the growing sub-communities in the NLP research community for each of these areas, such as the FinNLP workshop for financial technology and NLP at `https://sites.google.com/nlg.csie.ntu.edu.tw/finnlp-2022/`.